

1. Describe, with the aid of a diagram, the structural transformation of 1.2% carbon steel as it is cooled from austenite state.
 2. Using suitable examples for each, describe the following bonding mechanism:
 - (i) Metallic bond
 - (ii) Ionic bond
 - (iii) Covalent bond
 3. A bar with rectangular cross section (20mm x 40mm) and length 2.8m is subjected to an axial tensile force of 70KN. Calculate the tensile stress and strain in the bar if elongation is 1.2 mm.
 4. State any five reasons for testing materials.
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1. Distinguish between elastic and plastic deformation.
2. Describe, with aid of a diagram, the cooling of a pure metal.
3. Describe briefly the mechanism of fracture.
4. Enumerate five methods of controlling corrosion.
5. Describe the following micro-constituents of steel:
 - (i) Ferrite
 - (ii) ~~Au~~ Austenite
 - (iii) Pearlite
 - (iv) Cementite



KENYATTA UNIVERSITY
UNIVERSITY EXAMINATIONS 2020/2021
FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE (MECHANICAL ENGINEERING)
EMM 102: INTRODUCTION TO MATERIAL SCIENCE

DATE: TUESDAY 16TH MARCH 2021

TIME: 8.00 A.M. – 10.00 A.M.

INSTRUCTIONS:

Answer question ONE and any other TWO questions

1. (a) Describe, with the aid of a diagram, the structural transformation of 1.2% carbon steel as it is cooled from austenite state. (10 marks)
- (b) Using suitable examples for each, describe the following bonding mechanisms:
 - (i) Metallic bond
 - (ii) Ionic bond
 - (iii) Covalent bond (10 marks)
- (c) A bar with rectangular cross section (20mm × 40mm) and length 2.8m is subjected to an axial tensile force of 70 kN. Calculate the tensile stress and strain in the bar if elongation is 1.2mm. $\sigma = 0.875$, $\epsilon = 2.86 \times 10^{-3}$ (5 marks)
- (d) State any five reasons for testing materials. (5 marks)
2. (a) Give an outline of the mechanism of recrystallisation. (5 marks)
- (b) Distinguish between elastic and plastic deformation. (5 marks)
- (c) Which are the factors that affect the recrystallisation temperature? (6 marks)
- (d) Describe the difference between destructive and non-destructive testing. (4 marks)
3. (a) Describe the factors that affect grain growth. (6 marks)
- (b) Enumerate six methods of controlling corrosion. (6 marks)
- (c) Define the following:
 - (i) Eutectoid reaction
 - (ii) Peritectic reaction
 - (iii) Peritectoid reaction
 - (iv) Eutectic (8 marks)

4. (a) Describe the mechanism of dispersion strengthening. (5 marks)
- (b) List five types of protective coating techniques. (5 marks)
- (c) Describe briefly the mechanism of fracture. (5 marks)
- (d) Describe, with aid of a diagram, the cooling of a pure metal. (5 marks)

5. (a) Describe the following micro-constituents of steel:

- (i) Ferrite
- (ii) Austenite
- (iii) Pearlite
- (iv) Cementite

(10 marks)

(b) Describe, with aid of diagrams, the specimen and testing procedure for tensile testing.

What properties can be estimated using this test (5 marks)

(c) List five factors that influence the corrosion rate (5 marks)